Quantum Field Theory in a Nutshell – Anthony Zee

Notes by Gene

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Part I

Motivation and Foundation

We work with the God-given units $c = \hbar = 1$

1 Mathematical Preliminaries

1.1 Path-Integral Formalism

Consider a double-slit experiment. To get the total amplitude $\langle q_f | e^{-iHt/\hbar} | q_i \rangle$, we sum over the amplitude of each path. But what if we drill infinite holes in the barrier, so there's no barrier at all?

Then we must sum over **all possible paths.** This was Feynman's great observation, aided by Dirac's earlier work.

$$\mathcal{A}(x_i, x_f;\ dt) = \sum_{\text{paths}} \mathcal{A}(x_i, x_f;\ dt)$$

We divide time T into N segments of $\delta t = T/N$, sending $N \to \infty$. By remembering the resolution of

identity and $\langle q|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipq/\hbar}$, we eventually get

$$\langle q_f|e^{-iHt/\hbar}|q_i\rangle = \int \mathcal{D}[q(t)]\; e^{iS/\hbar}$$

$$\int \mathcal{D}[q(t)] = \lim_{N\to\infty} \left(\frac{-im}{2\pi\hbar\delta t}\right)^{N/2} \left(\prod_{k=1}^{N-1} \int dq_k\right)$$

$$S = \int_0^T L(q,\dot{q})dt$$

For algebraic details, read the book. I don't want to do it again.